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CS 621 DATA ANALYTICS

Text Analytics

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TEXT ANALYTICS

- After engineered the unstructured text data collected earlier, we can perform common text analytics tasks such as
  - Word Usage Analysis via word counts (Word Frequency Analysis)
  - Word Cloud for text visualization
  - Other types of text analytics and visualization

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WORD FREQUENCY ANALYSIS

- A dictionary is used to store the word frequencies as
  {“like”: 98, “love”:50, “dislike”:24}

- Pandas Series and DataFrame objects are frequently used in word frequency analysis.

- Several visualizations are commonly used to show results of word frequency analysis.

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WORD FREQUENCY COMPUTATION

- Compute a list of words, and a list of distinct words.
  Traverse two lists to count the frequency of each distinct word.
- Create a Series Object of words first.
  Apply .value_counts() to find out the count of each unique index word.
- Create a DataFrame object with one column of words first.
  Select the column and apply .value_counts().
- Consider saving the word-frequency result into a file at this stage.

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WORD CLOUD

- Word cloud is the most commonly used text visualization.

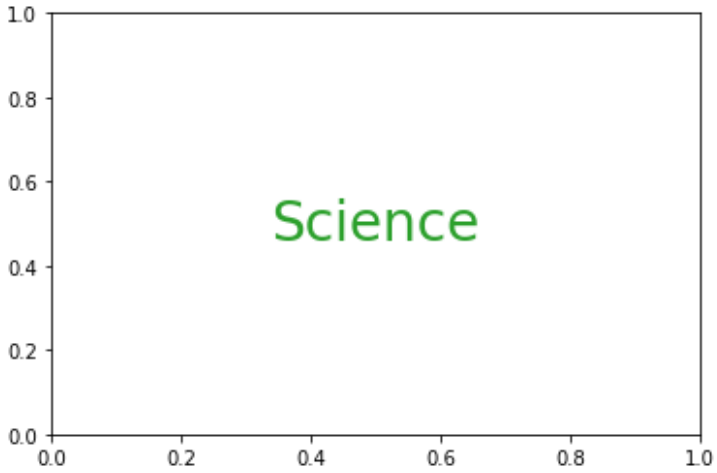


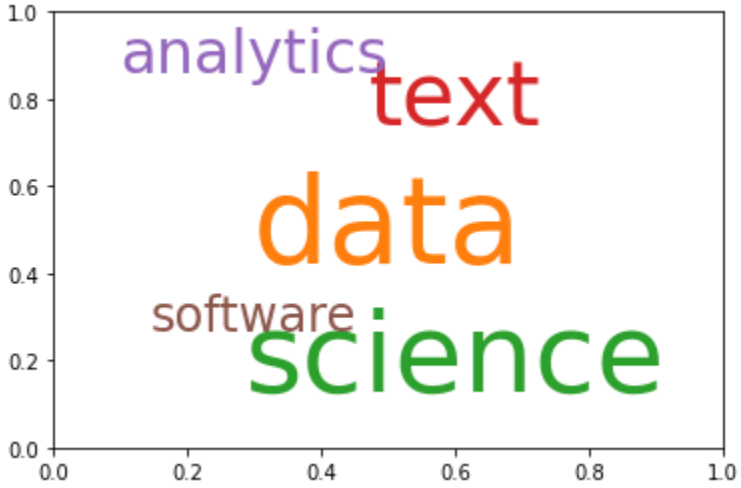
- Use wordcloud library, available from
  https://pypi.org/project/wordcloud/
- WordItOut is a web service for generating word cloud.

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PYPLOT.TEXT()

- We can draw text using matplotlib.pyplot.text(). Not simple to do, but help us understand how text visualization been done under the hood.





- See more details in class code samples.

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WORDCLOUD API

- >>> from wordcloud import WordCloud
- >>>WordCloud().generate(text)
- >>> WordCloud.generate_from_frequencies(freq)

- Many parameters available like:
  - styles
  - font size
  - font color
  - background color /background mask image
  - stopwords
  - max_words
- See class code demonstration for more.

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WORD CLOUD MASK

- Sometimes, it is fun to add a mask to the word cloud.



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POS ANALYSIS

- POS stands for Part of Speech.
- POS Analysis is important in language analysis.
- Perform POS tagging via language APIs.
- nltk.pos_tag(text)
- Common tags:
  - NN: Noun
  - NNP: Proper Noun singular
  - NNS: Noun, plural
  - VB?: Verb
  - JJ: Adjective
  - RB: Adverb
  - …

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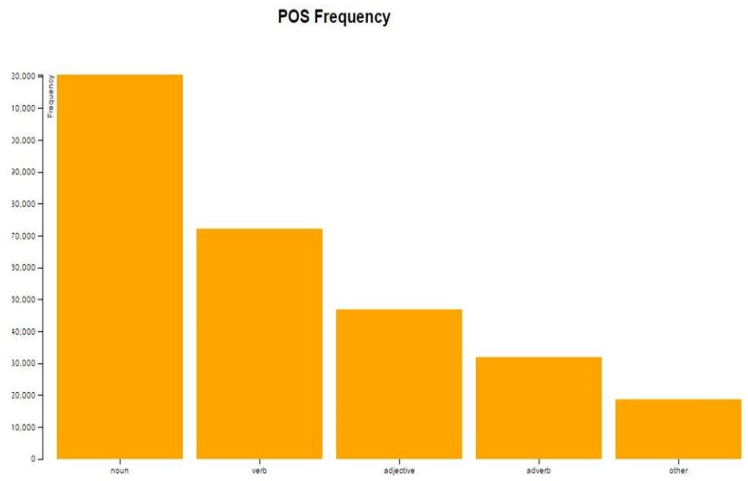
POS ANALYSIS

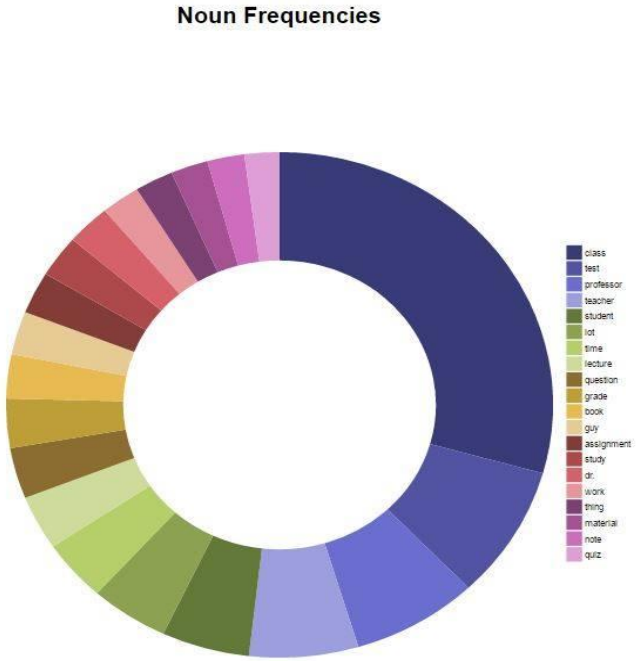
```
Data science is a great subject to study.
Software programming is fun.
I love both data and computer programming.
CS617 is programming for data science.
```

```
[[('Data', 'NNP'),
  ('science', 'NN'),
  ('is', 'VBZ'),
  ('a', 'DT'),
  ('great', 'JJ'),
  ('subject', 'NN'),
  ('to', 'TO'),
  ('study', 'NN'),
  ('.', '.')],
 [('Software', 'NNP'),
  ('programming', 'NN'),
  ('is', 'VBZ'),
  ('fun', 'NN'),
  ('.', '.')],
 [('I', 'PRP'),
  ('love', 'VBP'),
  ('both', 'DT'),
  ('data', 'NNS'),
```

OTHER TEXT VISUALIZATIONS

- Other types of visualizations like
  bar chart and pie chart are also often used.







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WE FLY